



GW Instek's GPI-745A electrical safety tester

riorating in its performance as an insulator (extremely bad conductor) with time. The most important reason for testing insulation is to ensure public and personal safety. "Insulation testing is used to determine the integrity of windings or cables in motors, transformers switchgear and electrical installations," shares Prachi Gupta, director, Avante Global. "By performing a high-potential test between de-energised current-carrying, grounded and grounding conductors, you can eliminate the possibility of having a life-threatening short-circuit or short-to-ground faults," she adds.

Electrical and electronic systems when exposed to environmental factors such as dirt, grease, temperature, stress and vibration affect the dielectric material used in insulation. This may lead to insulation failure, resulting in loss of production or even fire. "It is important to identify the deterioration as early as possible to take corrective measures," suggests Ms Wang.

HIPOT tests

When a high potential is applied between isolated parts of a circuit or a product, the behaviour of electrical parameters helps identify broken or poor insulation, stray wire strands or braided shielding, spacing problem between terminals, tolerance errors, etc. This helps in making informed decision on the safety and quality of electrical circuits.

"The value/range of HIPOT de-

pends on the device under test (DUT)," shares Rohit Manwani, chief executive officer of Sunil Associates. "Range of the instrument varies from industry to industry. Some industries can accept the sample if insulation resistance is more than 1000 mega-ohms, while some look for values in giga-ohms," he adds.

High-potential tests can be divided into three distinct tests with different goals. To understand the high-voltage capacity of an insulating material, dielectric breakdown test is used. The voltage between the test points is in-

creased continuously until the current increases suddenly. This is the point where insulation gets into conduction. Normally, this test is done during the product design and qualification stage.

"Of course, the sample DUT that fails the test is rejected as it is completely damaged now," shares Manwani. The dielectric withstand test is normally done at 75 per cent of the breakdown voltage. The test is performed to understand for how long the dielectric can withstand high voltage.

Insulation resistance (IR) test measures resistance of the insulating material. You apply a very high voltage across the dielectric, carefully measure the current and calculate the resistance using Ohms law. "For most of the products, the test is carried at 500V DC and the resistance should be at least 2 mega-ohms," shares Ms Wang. "Usually IR is checked first, before putting the device for HIPOT test," she adds.

HIPOT testing can be carried by an AC or DC voltage. However, there are different considerations when using AC or DC voltages. The difference in the reading becomes significant when the DUT is highly capacitive. "AC volt-

Types of leakage currents

During the insulation test, a high voltage is generated that causes a small current (in micro-amperes) to flow through the conductor and insulation. The amount of current depends on the amount of voltage applied, the system's capacitance, total resistance and temperature of the material. The value of insulation resistance read on the meter will be a function of following three independent sub-currents:

Conductive leakage current. Conductive current is a small current that normally flows through insulation, between conductors or from a conductor to ground. This current increases as insulation deteriorates and becomes predominant after the absorption current vanishes.

Capacitive charging leakage current. When two or more conductors are run together in a raceway, these act as a capacitor. Due to this capacitive effect, a leakage current flows through conductor insulation. This current lasts only a few seconds as the DC voltage is applied and drops out after the insulation has been charged to its full test voltage. In low-capacitance equipment, capacitive current is higher than conductive leakage current, but usually disappears by the time you start recording the data. Because of this, it is important to let the reading settle out before recording it. On the other hand, when testing high-capacitance equipment, the capacitive charging leakage current can last a very long time before settling out.

Polarisation absorption leakage current. Absorption current is caused by polarisation of molecules within the dielectric material. In low-capacitance equipment, current is high for the first few seconds and decreases slowly to nearly zero. When dealing with high-capacitance equipment or wet and contaminated insulation, there will be no decrease in the absorption current for a long time.

—Prachi Gupta, director, Avante Global